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Improving Physical Layer Security Performance for Dual-Hop Networks Using ZFB and NAN

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DECLARATION

I, **Druscilla Dorcus Namata**, do hereby declare that this project report is my own original work and has not been submitted, either in whole or in part, for a degree or any other academic award at Makerere University or any other institution.

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ABSTRACT

The rapid proliferation of fifth-generation (5G) and beyond-5G wireless networks, alongside the exponential growth of Internet of Things (IoT) deployments, has made improving the security of wireless transmissions a paramount engineering concern. The inherently broadcast nature of wireless channels renders them vulnerable to passive eavesdropping, a threat that is particularly acute in resource-constrained dual-hop networks where conventional cryptographic mechanisms impose prohibitive computational overhead. This report focuses on improving physical layer security (PLS) performance through the implementation and comparative analysis of advanced cooperative jamming strategies.

The study investigates a dual-hop amplify-and-forward (AF) cooperative wireless network comprising a single-antenna base station (BS), a multi-antenna relay, a single-antenna legitimate destination (Bob), and a passive single-antenna eavesdropper (Eve). To reflect the propagation heterogeneity observed in practical IoT deployments, the system model incorporates Rayleigh fading environments across all links. Performance improvement is evaluated through two primary strategies: Transmit Antenna Selection with Maximal Ratio Combining using Zero-Forcing Beamforming (TAS-MRC-ZFB), and a similar diversity framework utilizing Null-Space Artificial Noise (TAS-MRC-NAN).

A key contribution of this work is the analysis of performance improvements in scenarios where the eavesdropper's channel state information (CSI) is unavailable, necessitating a reliance on the null-space structure of the legitimate channel. To provide a rigorous evaluation of these improvements, exact closed-form expressions for the secrecy outage probability (SOP) and effective secrecy throughput (EST) are derived using order statistics and incomplete gamma function theory.

The results reveal that TAS-MRC-ZFB provides a significantly superior improvement in security performance compared to TAS-MRC-NAN. Specifically, the ZFB strategy achieves up to two orders of magnitude lower SOP at high SNR regimes and sustains a higher peak EST, reaching approximately 1.5 bps/Hz for a five-antenna relay configuration. These findings confirm that Zero-Forcing Beamforming is the preferred mechanism for improving PLS performance in cooperative networks, providing actionable design guidance for secure sensor networks in heterogeneous fading environments.

Keywords: Physical Layer Security; Cooperative Jamming; Rayleigh Fading; Secrecy Outage Probability; Zero-Forcing Beamforming; Null-Space Artificial Noise; Transmit Antenna Selection; Maximal Ratio Combining; Amplify-and-Forward Relay.

DEDICATION

To my family, whose unwavering support and sacrifices made this journey possible.

*To my project partner, **Isabella Kasigwa** who has been by my side throughout this project time.*

To my mentors and lecturers at Makerere University, who ignited in us a passion for communications engineering and guided us with patience and wisdom.

And to the broader scientific community striving to build secure, trustworthy wireless systems for the benefit of humanity.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
5G	Fifth Generation (wireless network standard)
AF	Amplify-and-Forward
AN	Artificial Noise
AWGN	Additive White Gaussian Noise
BS	Base Station
CDF	Cumulative Distribution Function
CJ	Cooperative Jamming
CSI	Channel State Information
DF	Decode-and-Forward
EST	Effective Secrecy Throughput
i.i.d.	Independent and Identically Distributed
IoT	Internet of Things
MRC	Maximal Ratio Combining
NAN	Null-Space Artificial Noise
PDF	Probability Density Function
PLS	Physical Layer Security
RIS	Reconfigurable Intelligent Surface
SNR	Signal-to-Noise Ratio
SOP	Secrecy Outage Probability
SWIPT	Simultaneous Wireless Information and Power Transfer
TAS	Transmit Antenna Selection
TAS-MRC-NAN	Transmit Antenna Selection–Maximal Ratio Combining–Null-Space Artificial Noise
TAS-MRC-ZFB	Transmit Antenna Selection–Maximal Ratio Combining–Zero-Forcing Beamforming
WSN	Wireless Sensor Network
ZFB	Zero-Forcing Beamforming

LIST OF PRINCIPAL SYMBOLS

Symbol	Description
P_A	Transmit power of the base station (source node)
P_R	Transmit power of the relay
P_J	Jamming signal power
N_0	Additive white Gaussian noise variance
M	Number of antennas at the relay
α	Path-loss exponent
d_{ij}	Distance between node i and node j
$\mathbf{h}_{AR}$	Channel vector from BS (Alice) to Relay ($M \times 1$)
h_{AE}	Channel coefficient from BS to Eve
$\mathbf{h}_{RB}$	Channel vector from Relay to Bob ($M \times 1$)
$\mathbf{h}_{RE}$	Channel vector from Relay to Eve ($M \times 1$)
$\bar{\gamma}_{AR}$	Average SNR from BS to Relay
$\bar{\gamma}_{RB}$	Average SNR from Relay to Bob
$\bar{\gamma}_{AE}$	Average SNR from BS to Eve
$\bar{\gamma}_{RE}$	Average SNR from Relay to Eve
γ_B	Instantaneous SNR at the legitimate destination (Bob)
γ_E	Instantaneous SNR at the eavesdropper (Eve)
γ_{ARB}	End-to-end instantaneous SNR through the dual-hop relay
γ_{AR}	Instantaneous SNR from BS to Relay
γ_{RB}^{TAS}	Instantaneous SNR of TAS-selected relay-to-Bob link
$\mathbf{w}$	MRC receive beamforming vector at relay
$\mathbf{w}_2$	Relay transmit beamforming vector
G	Relay amplification gain factor
$\mathbf{z}$	Artificial noise vector (NAN scheme)
C_s	Achievable secrecy capacity (bps/Hz)
R_s	Target secrecy rate threshold (bps/Hz)
$P_{\text{out}}(R_s)$	Secrecy outage probability
$\mathcal{ST}(R_s)$	Effective secrecy throughput (bps/Hz)
$F_\gamma(\cdot)$	Cumulative distribution function of random variable γ
$f_\gamma(\cdot)$	Probability density function of random variable γ
$\Gamma(\cdot)$	Gamma function
$\Gamma(a, x)$	Upper incomplete gamma function
Ω_{ij}	Mean channel power gain $\mathbb{E}\{ h_{ij} ^2\}$
$\dagger$	Conjugate transpose operator
$\ \cdot\ _F$	Frobenius norm
$\mathbf{I}$	Identity matrix

Symbol	Description
A, B	Secrecy threshold constants: $A = 2^{R_s} - 1$, $B = 2^{R_s}$
$\alpha_{\text{ZFB}}, \alpha_{\text{NAN}}$	Eavesdropper SNR ratio parameters
$\beta_2, \beta_3, \beta_4$	Composite exponent parameters in SOP expressions
$\Phi(p, q, \beta, \alpha)$	Auxiliary function in NAN SOP derivation

CHAPTER 1: INTRODUCTION

1.1 Background

The global telecommunications landscape is undergoing an unprecedented transformation. The deployment of fifth-generation (5G) networks, combined with the explosive growth of the Internet of Things (IoT), has created a hyperconnected world in which billions of devices exchange sensitive data wirelessly. By 2025, it was estimated that over 75 billion IoT devices were connected to the internet, spanning applications from industrial automation and smart grids to remote healthcare monitoring, autonomous transportation systems, and intelligent surveillance networks [1, 2]. These mission-critical applications require not only reliable communication but also strong guarantees of data confidentiality.

Wireless communication, however, is inherently insecure by its very nature. Unlike wired transmission, electromagnetic signals propagate through open space and can be received by any party within range, not just the intended recipient. This broadcast characteristic exposes wireless transmissions to passive eavesdropping, a threat that is, by design, undetectable: an eavesdropper can intercept a communication without transmitting any signal and without alerting the legitimate parties [3]. As wireless networks extend into critical infrastructure, the consequences of such interceptions can be severe, ranging from industrial espionage and theft of financial data to disruption of safety-critical systems.

The conventional response to this threat has been cryptographic security. Techniques such as public-key encryption, digital signatures, and symmetric key protocols form the bedrock of modern network security. In resource-rich environments with powerful processors and stable key management infrastructure, these approaches are highly effective. However, the constraints of modern IoT deployments challenge their viability. Lightweight sensor nodes often possess limited processing power and memory, making the computational overhead of key-exchange protocols prohibitive [6]. Battery-powered nodes must minimise energy expenditure, and the cost of running complex encryption algorithms is non-trivial. Furthermore, the emerging threat of quantum computing places the long-term integrity of many classical cryptographic primitives in serious jeopardy [7]. Algorithms such as RSA and elliptic-curve cryptography, which underpin much of today's secure communications, could be broken by sufficiently powerful quantum computers using Shor's algorithm, necessitating the development of alternative security paradigms.

Physical layer security (PLS) has emerged over the past two decades as a mathematically rigorous and hardware-compatible alternative to cryptographic approaches [8]. Rather than securing communications through computational com-

plexity, PLS exploits the inherent randomness and spatial structure of wireless fading channels to achieve information-theoretic secrecy. The foundational insight, provided by Shannon’s information-theoretic framework [7] and formalised by Wyner’s wiretap channel model [9], is that a positive rate of perfectly secret information transmission, the secrecy capacity, is achievable whenever the legitimate receiver’s channel is statistically stronger than the eavesdropper’s channel, without any reliance on shared cryptographic keys. This is a powerful result because it shifts the security guarantee from computational assumptions to the fundamental laws of physics and probability.

1.1.1 Cooperative Relay Networks and Their Security Challenges

In practice, the gap between the legitimate channel and the eavesdropper’s channel cannot always be guaranteed. In scenarios where the legitimate receiver is far from the transmitter or is subject to deep fading, while the eavesdropper is positioned closer to the transmitter or enjoys better propagation conditions, the instantaneous secrecy capacity can drop to zero [10]. This fundamental vulnerability of single-hop PLS has motivated extensive research into cooperative relay networks as a means of enhancing physical layer security.

In a cooperative relay network, one or more intermediate relay nodes assist in the delivery of information from the source to the destination. Amplify-and-forward (AF) relaying, in which the relay simply amplifies and retransmits the received signal without decoding, is particularly attractive for low-complexity IoT deployments. By creating a two-hop path from the base station to the legitimate user, AF cooperative networks can improve the signal quality at the destination and extend coverage. However, the relay itself introduces new security considerations: the relay’s transmissions may be intercepted by an eavesdropper positioned along the relay-to-destination link.

1.1.2 Cooperative Jamming and Antenna Diversity

One of the most effective strategies for enhancing PLS in cooperative networks is cooperative jamming (CJ), in which the relay deliberately transmits an interference signal alongside the information signal to degrade the quality of reception at the eavesdropper [13, 14]. When the relay is equipped with multiple antennas, sophisticated spatial processing techniques can be employed to ensure that the jamming signal does not impair the legitimate receiver while maximally disrupting the eavesdropper’s reception. Two principal approaches exist for designing the relay’s transmit beamforming vector: Zero-Forcing Beamforming (ZFB), which projects the information signal into the null space of the eavesdropper’s channel to completely eliminate the relay-to-eavesdropper information leakage; and Null-Space Artificial Noise (NAN), which injects noise into the null space of the legitimate user’s channel to ensure that no interference is added to the desired signal.

An important practical distinction between the two approaches concerns the channel state information (CSI) they require. ZFB requires knowledge of the eavesdropper’s channel vector at the relay, which is unavailable in passive eavesdrop-

ping scenarios. NAN, by contrast, requires only knowledge of the legitimate user's channel, which is routinely available through standard channel estimation and feedback mechanisms. However, in certain network geometries and with sufficient relay antenna count, ZFB can achieve substantially better secrecy performance, making it the preferred choice when the eavesdropper's approximate location or statistical channel model can be estimated.

Transmit Antenna Selection (TAS) combined with Maximal Ratio Combining (MRC) is a well-established diversity technique that achieves full receive diversity with a single radio-frequency (RF) chain at the relay, minimising hardware complexity while maximising signal quality at the legitimate receiver [19]. The relay selects the single transmit antenna that maximises the signal-to-noise ratio (SNR) at the destination, while MRC at the receiver combines all received signal copies coherently. This combination of TAS at the transmitter and MRC at the receiver represents an attractive balance of performance and practical feasibility.

1.1.3 Rayleigh Fading Environments

Real-world wireless networks are characterised by rich multipath propagation environments. The Rayleigh distribution is the standard statistical model for wireless fading in environments without a dominant line-of-sight component, and accurately captures the behaviour of mobile and sensor network links in urban and indoor deployments where scattering is rich [5]. The Rayleigh fading model is adopted uniformly across all links in this work, providing a tractable and widely accepted analytical framework for the study of physical layer security in cooperative IoT sensor networks [12, 18].

1.2 Problem Statement

Despite the substantial body of literature on physical layer security, several important gaps remain unaddressed in the context of practical IoT-oriented cooperative networks. Existing cooperative jamming studies have been largely confined to network configurations that do not simultaneously address the comparative performance of ZFB and NAN jamming strategies within a unified TAS-MRC framework. Furthermore, this comparative analysis has not been rigorously conducted with simultaneous consideration of eavesdropper proximity, relay antenna count, and the absence of eavesdropper CSI.

Specifically, the following technical challenges remain unresolved:

- i. No exact closed-form secrecy outage probability expression has been derived for a dual-hop AF TAS-MRC network under Rayleigh fading with cooperative jamming, for either ZFB or NAN strategies.
- ii. The quantitative performance gap between ZFB and NAN in terms of SOP and effective secrecy throughput under varying relay antenna count, eavesdropper SNR, and eavesdropper distance has not been systematically established.
- iii. The robustness of each jamming scheme to eavesdropper proximity to the

relay, a critical concern in passive eavesdropping scenarios where adversary placement cannot be controlled, has not been quantified.

- iv. The conditions under which NAN may serve as a viable alternative to ZFB, particularly when hardware or CSI constraints preclude zero-forcing, have not been identified through rigorous analysis.

These gaps motivate the analytical and simulation-based investigation conducted in this report.

1.3 Research Objectives

1.3.1 Main Objective of the Study

To improve physical layer security performance for dual-hop networks using ZFB and NAN.

1.3.2 Specific Objectives

- i. To design a system model for a cooperative network under Rayleigh and distance based large scale fading in cooperating TAS-MRC-ZFB and TAS-MRC-NAN strategies with no eavesdropper CSI at the relay.
- ii. To derive close-form expressions for the key performance metrics such as secrecy outage probability, effective secrecy throughput.
- iii. To conduct a cooperative performance analysis through numerical results validated by analytical simulations benchmarking both schemes across key parameters.

1.4 Scope of the Project

This project focuses on a specific network topology consisting of a single-antenna base station, an M -antenna AF relay, a single-antenna legitimate destination, and a single-antenna eavesdropper. The following assumptions and limitations define the scope of the analysis:

- The direct link from the base station to the legitimate destination is assumed to be blocked (e.g., by a building or obstacle), so all communication occurs exclusively via the relay. This models practical urban or indoor scenarios where line-of-sight is unavailable.
- The eavesdropper has a direct link from the base station and also receives the relay's transmission. This represents a conservative and practically relevant threat model.
- The relay has perfect knowledge of the legitimate user's channel state information, but no knowledge of the eavesdropper's channel state information, which is the typical constraint in passive eavesdropping scenarios.

- Small-scale fading is modelled as Rayleigh for all links. Large-scale path loss is modelled with exponent α .
- Thermal noise is modelled as additive white Gaussian noise (AWGN) with variance N_0 at all nodes.
- The analysis is confined to the secrecy outage probability and effective secrecy throughput as performance metrics. Power optimisation and threshold-based mode switching are identified as areas for future work.
- MatLab simulation results are used for validation rather than as an independent design tool.

1.5 Significance of the Study

The significance of this research extends across theoretical, practical, and policy dimensions:

Theoretical Contribution: This work provides, for the first time, exact closed-form analytical expressions for the SOP of dual-hop AF TAS-MRC cooperative networks under Rayleigh fading with cooperative jamming. These results advance the mathematical theory of physical layer security and provide a rigorous benchmark against which future schemes can be evaluated.

Practical Relevance: The network model and system assumptions are directly motivated by real IoT sensor network deployments, where passive eavesdropping threats and hardware constraints on relay nodes are the norm rather than the exception. The design insights derived from this analysis provide engineers with quantitative guidance on when ZFB is worth the complexity overhead of eavesdropper channel estimation, and when NAN may suffice.

Security Engineering: As 5G networks are rolled out across Uganda and sub-Saharan Africa, securing the wireless infrastructure against passive interception attacks becomes increasingly important. This research contributes to the knowledge base for designing secure wireless networks in environments characterised by resource-constrained nodes, directly relevant to the emerging connected infrastructure of the region.

1.6 Report Organisation

The remainder of this report is organised as follows. Chapter 2 presents a comprehensive review of the relevant literature, tracing the development of physical layer security theory from its information-theoretic foundations through cooperative relaying, cooperative jamming, and antenna diversity strategies. Chapter 3 describes the proposed system model and derives the end-to-end SNR expressions for both the TAS-MRC-ZFB and TAS-MRC-NAN schemes. Chapter 4 presents the secrecy performance analysis, deriving exact closed-form expressions for the SOP and EST of both schemes. Chapter 5 presents the numerical results and simulation validation, with a systematic comparative performance analysis. Chapter 6 draws conclusions and identifies directions for future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The field of physical layer security sits at the intersection of information theory, signal processing, and wireless communications engineering. This chapter reviews the key theoretical developments and recent advances that form the intellectual foundation of this project. The review is organised thematically, progressing from the foundational information-theoretic results through cooperative relay security, cooperative jamming strategies, antenna diversity for PLS, and fading channel models.

2.2 Foundations of Physical Layer Security

2.2.1 Shannon's Secrecy Framework

The theoretical basis for physical layer security traces directly to Shannon's landmark 1949 paper on the communication theory of secrecy systems [7]. Shannon demonstrated that perfect secrecy, the condition in which an adversary gains absolutely no information from an intercepted ciphertext, is achievable if and only if the entropy of the secret key is at least as large as the entropy of the message. While this result motivated the development of one-time pads, it also highlighted the fundamental information-theoretic constraints on secret communication.

Shannon's framework was, however, primarily concerned with symmetric key cryptography and did not directly address the physical channel. The pivotal step toward exploiting the physical channel for security was taken by Wyner in his seminal 1975 paper introducing the degraded wiretap channel [9]. Wyner demonstrated that if the eavesdropper's channel is a statistically degraded version of the legitimate receiver's channel, meaning the eavesdropper receives a noisier version of what the legitimate receiver sees, then a positive secrecy rate is achievable without any shared key. The maximum achievable rate of perfectly secret communication, termed the secrecy capacity C_s , is defined as $C_s = [C_B - C_E]^+$, where C_B and C_E are the channel capacities of the legitimate and eavesdropper channels, respectively, and $[\cdot]^+ = \max(\cdot, 0)$ denotes the positive part.

Csiszár and Körner subsequently generalised Wyner's result to broadcast channels with a confidential message [5], establishing the secrecy capacity for non-degraded wiretap channels. Leung-Yan-Cheong and Hellman extended the analysis to Gaussian channels, showing that the secrecy capacity of the Gaussian wiretap channel is simply the difference of the channel capacities of the two links when the eavesdropper's channel is weaker. These foundational results established PLS as a rigorous information-theoretic discipline with concrete performance guarantees.

2.2.2 Secrecy Outage Probability in Fading Channels

In quasi-static fading channels, the instantaneous channel gains vary randomly due to multipath propagation and mobility, so the secrecy capacity itself becomes a random variable. A deterministic positive secrecy rate cannot be guaranteed for all channel realisations. Instead, the secrecy outage probability (SOP) has become the principal performance metric for PLS in fading channels [3]. The SOP is defined as the probability that the instantaneous secrecy capacity falls below a prescribed target secrecy rate R_s :

$$P_{\text{out}}(R_s) = \Pr(C_s < R_s). \quad (2.1)$$

A low SOP corresponds to a high probability of maintaining the target secrecy rate, i.e., reliable secure communication. Studies under Rayleigh fading have established that the SOP decreases monotonically with the legitimate user's SNR and exhibits a diversity order equal to the number of independently fading paths in the legitimate channel [10].

2.2.3 Effective Secrecy Throughput

While the SOP captures the probability of maintaining a target secrecy rate, it does not fully characterise the trade-off between secrecy reliability and information efficiency. The effective secrecy throughput (EST) addresses this limitation by measuring the average rate of successfully transmitted secret information:

$$\mathcal{T}(R_s) = (1 - P_{\text{out}}(R_s)) \cdot R_s. \quad (2.2)$$

The EST is a concave function of R_s : for small R_s , the SOP is nearly zero and $\mathcal{ST} \approx R_s$; as R_s increases, the SOP rises until it dominates and $\mathcal{ST}$ decreases. The optimal secrecy rate R_s^* that maximises the EST provides a practically meaningful operating point for secure system design [16].

2.3 Physical Layer Security in Cooperative Relay Networks

2.3.1 Motivation for Cooperative Relaying

The secrecy capacity of a single-hop link depends critically on the relative channel quality of the legitimate and eavesdropper links. When the eavesdropper is in an advantageous position, closer to the transmitter than the legitimate receiver, for instance, the secrecy capacity may collapse to zero for sustained periods. Cooperative relay networks offer a powerful means of overcoming this limitation by creating an additional communication path that can bypass the eavesdropper's advantageous position or degrade its reception through interference.

Two principal relay protocols have been extensively studied in the PLS context: decode-and-forward (DF), in which the relay fully decodes the received signal and re-encodes it for retransmission, and amplify-and-forward (AF), in which the relay simply scales and retransmits the received signal without decoding. While DF achieves better noise performance, AF is preferred in low-complexity or energy-constrained applications because it does not require full decoding at the relay [13]. Both protocols have been rigorously analysed under various relay selection criteria, with results confirming that optimal relay selection achieves full secrecy diversity order equal to the number of available relay nodes.

2.3.2 Secrecy Performance of AF Relay Networks

Ikki and Aissa established performance bounds for dual-hop AF relay networks in the presence of co-channel interference [21], showing that the end-to-end SNR can be accurately approximated by the minimum of the two-hop SNRs in the high-SNR regime. This tight approximation greatly simplifies the analysis of dual-hop systems and is widely used in the PLS literature [22]. Huang et al. derived closed-form secrecy expressions for multi-antenna cooperative relay networks with TAS, confirming that the secrecy diversity order scales with the number of relay antennas [23].

Under Rayleigh fading, the SOP decreases monotonically with transmit SNR and the achievable diversity order is determined by the number of relay antennas and the TAS-MRC combining strategy [19]. Extensions to multi-relay topologies, imperfect CSI scenarios, and full-duplex relay architectures have further demonstrated the versatility of cooperative networks as platforms for PLS enhancement [13, 14].

2.3.3 Relay Selection for PLS

Krikidis, Thompson, and McLaughlin studied relay selection strategies for secure cooperative networks with jamming [14], demonstrating that joint relay selection and power allocation can substantially reduce the SOP compared to random relay selection. This result established the importance of exploiting multi-relay diversity for security, not merely for reliability. Subsequent work on threshold-based mode switching showed that activating cooperative jamming only when the relay-to-legitimate-user channel exceeds a threshold provides significant SOP reductions compared to always-on jamming [10].

2.4 Cooperative Jamming Strategies

2.4.1 Artificial Noise Injection

The foundational artificial noise (AN) framework was established by Goel and Negi [15], who demonstrated that a multi-antenna transmitter can project noise into the null space of the legitimate channel, guaranteeing secrecy without eavesdropper CSI. The key insight is that the null space of the legitimate channel is orthogonal to the desired signal vector: noise projected onto this space adds interference at the eavesdropper without degrading reception at the legitimate receiver. This seminal idea has since spawned an extensive literature on AN-based PLS.

The artificial noise approach was subsequently extended to cooperative relay networks, where the relay acts as a jammer to degrade the eavesdropper's channel during the second transmission phase. Yang et al. studied secure multiuser communications in wireless sensor networks with TAS and cooperative jamming [24], demonstrating that the combination of TAS for legitimate signal transmission and AN injection for jamming achieves attractive secrecy performance with practical receiver complexity. This work is particularly relevant to the present analysis as it establishes the theoretical framework for TAS-MRC with null-space jamming.

2.4.2 Zero-Forcing Beamforming

Zero-forcing beamforming (ZFB) takes a different approach to cooperative jamming: instead of injecting noise into the null space of the legitimate channel, ZFB projects the

transmitted signal itself into the null space of the eavesdropper's channel, completely eliminating the relay-to-eavesdropper information leakage [24]. This is achieved by designing the relay's beamforming vector as the normalised projection of the relay-to-Bob channel vector onto the complement of the relay-to-Eve channel space. The resulting scheme achieves a theoretical secrecy advantage because the eavesdropper receives no relay signal at all, relying only on the direct BS-to-Eve link. The trade-off is that ZFB requires knowledge of the eavesdropper's channel vector at the relay, which is unavailable in purely passive eavesdropping scenarios. In practice, this limitation may be partially addressed through statistical channel models or by estimating the eavesdropper's location.

Wang, Zhou, and Chen studied RIS-aided secure transmission for energy-constrained IoT networks with imperfect CSI [17], highlighting that ZFB's advantage over NAN is most pronounced when the number of relay antennas is large, providing additional spatial degrees of freedom for null-space projection. Zou, Zhu, Wang, and Hanzo surveyed wireless security techniques including ZFB and NAN in [16], identifying ZFB as the preferred strategy when eavesdropper CSI is available, but noting that NAN's simpler CSI requirement often makes it more practically appealing.

2.4.3 Null-Space Artificial Noise

Null-space artificial noise (NAN) represents the practically preferred cooperative jamming strategy in passive eavesdropping scenarios [15]. The relay transmits the information signal using TAS to maximise the SNR at Bob, while simultaneously injecting artificial noise in the null space of the relay-to-Bob channel, ensuring no interference to the legitimate receiver. The noise, however, falls on the eavesdropper's channel with full power, degrading its SNR. The effectiveness of NAN decreases as the number of relay antennas increases, because the null space of Bob's channel becomes higher-dimensional and the interference power at the eavesdropper is diluted [24]. This creates an interesting trade-off: while TAS diversity improves with antenna count, the jamming effectiveness of NAN diminishes, leading to a performance gap relative to ZFB that widens with M .

2.5 Transmit Antenna Selection and Maximal Ratio Combining

2.5.1 TAS-MRC for Physical Layer Security

Transmit Antenna Selection combined with Maximal Ratio Combining (TAS-MRC) achieves full receive diversity with a single RF chain at the transmitter, requiring only the index of the best transmit antenna to be fed back from the receiver [19]. In the context of PLS, TAS-MRC is attractive because it improves the legitimate channel quality without requiring complex precoding or high-dimensional CSI feedback. Yang, Yeoh, El Kashlan, Schober, and Collings established that TAS-MRC achieves the same secrecy diversity order as optimal transmit precoding in MIMO wiretap channels [19], making it a near-optimal strategy with much reduced complexity.

Under Rayleigh fading, the secrecy diversity order of TAS-MRC scales linearly with the number of selected antennas [6]. The integration of TAS-MRC into cooperative jamming frameworks, specifically in combination with ZFB or NAN at the relay, represents the natural extension of these results to the dual-hop cooperative context, which is the

primary contribution of this project.

2.5.2 Applications in Sensor Networks

Ssettumba, Kasamba, Mutaawe, and Serugunda demonstrated the effectiveness of TAS-based PHY security improvements in wireless sensor networks in the presence of untrusted users [20], establishing a practical framework for the relay-based security enhancement studied in this report. Their work, conducted in the context of a Ugandan telecommunications research programme, motivated the extension of TAS-based security to the more general cooperative jamming framework analysed here. The present project builds directly on this foundation, extending the analysis to dual-hop AF relay networks with ZFB and NAN cooperative jamming under Rayleigh fading.

2.6 Rayleigh Fading Models for Wireless Networks

2.6.1 The Rayleigh Fading Distribution

The Rayleigh distribution is the standard model for wireless fading in environments characterised by rich multipath scattering and no dominant line-of-sight component [5]. It arises naturally when the received signal is the superposition of a large number of independently scattered components, each of random amplitude and phase. The squared magnitude of a Rayleigh-faded channel coefficient is exponentially distributed, which greatly facilitates closed-form analytical derivations in the PLS literature.

In IoT sensor networks, the Rayleigh model is appropriate for links between nodes deployed in urban environments, indoor spaces, or other rich-scattering settings where no single propagation path dominates. All links in the network considered in this project, namely the BS-to-relay, relay-to-Bob, relay-to-Eve, and BS-to-Eve links, are modelled as Rayleigh faded, reflecting the typical propagation conditions of practical wireless sensor network deployments [24].

2.6.2 Secrecy Analysis Under Rayleigh Fading

Under Rayleigh fading, the SOP of cooperative relay networks has been extensively studied, with results establishing its dependence on the SNR of each link, the diversity order of the combining strategy, and the eavesdropper's channel quality [12, 18]. The exponential distribution of Rayleigh-faded channel power gains enables the derivation of exact closed-form SOP expressions using tools from order statistics and the theory of the incomplete gamma function. These analytical techniques are directly employed in this project for the derivation of the TAS-MRC-ZFB and TAS-MRC-NAN SOP expressions.

2.7 Research Gaps and Motivation

A systematic review of the existing literature reveals the following research gaps that motivate this project:

- i. **No unified TAS-MRC framework with ZFB and NAN under Rayleigh fading:** Existing cooperative jamming studies have not jointly addressed TAS-

MRC cooperative relay networks with both multi-antenna ZFB and NAN jamming in a unified analytical framework [3, 5].

- ii. **No exact closed-form SOP for this configuration:** Closed-form SOP expressions for dual-hop AF TAS-MRC networks with cooperative jamming have not previously been derived, limiting design insight and practical applicability.
- iii. **Incomplete comparative analysis:** A rigorous comparative study of ZFB and NAN performance under the practical constraints of passive eavesdropping (no eavesdropper CSI at any node) has not been conducted, leaving engineers without quantitative guidance on strategy selection.
- iv. **Eavesdropper robustness not characterised:** The robustness of each jamming scheme to varying eavesdropper proximity to the relay – a critical concern in uncontrolled deployment environments – has not been studied systematically.

This project addresses all four gaps through the derivation of exact closed-form SOP and EST expressions, validated Monte Carlo simulations, and a comprehensive comparative performance analysis. The following chapter presents the system model that serves as the foundation for this analysis.

CHAPTER 3: SYSTEM MODEL AND METHODOLOGY

3.1 Introduction

This chapter presents the proposed system model, the channel and signal models for each transmission phase, and the derivation of the end-to-end SNR expressions at both the legitimate receiver and the eavesdropper under the TAS-MRC-ZFB and TAS-MRC-NAN cooperative jamming strategies. The chapter concludes with the definition of the secrecy capacity metric that forms the basis for the performance analysis in Chapter 4.

3.2 Network Topology

The proposed cooperative wireless network consists of four nodes:

- **Alice (BS):** A base station equipped with a single antenna, acting as the information source.
- **Relay (R):** An amplify-and-forward relay equipped with M antennas, performing both information forwarding and cooperative jamming.
- **Bob (B):** The legitimate destination, equipped with a single antenna.
- **Eve (E):** A passive eavesdropper with a single antenna, attempting to intercept the confidential transmission.

The direct link from Alice to Bob is assumed to be obstructed (e.g., by a building), making the relay the sole viable communication route from Alice to Bob. This is a practical assumption in urban deployment scenarios and focuses the analysis on the two-hop relay path. Eve is assumed to be unobstructed and proximate to Alice, providing her with both a direct link from Alice and the ability to intercept the relay's transmission. A schematic of the system model is illustrated conceptually in Figure 3.1. Alice transmits to the relay in Phase 1, and the relay forwards to Bob while simultaneously jamming Eve in Phase 2.

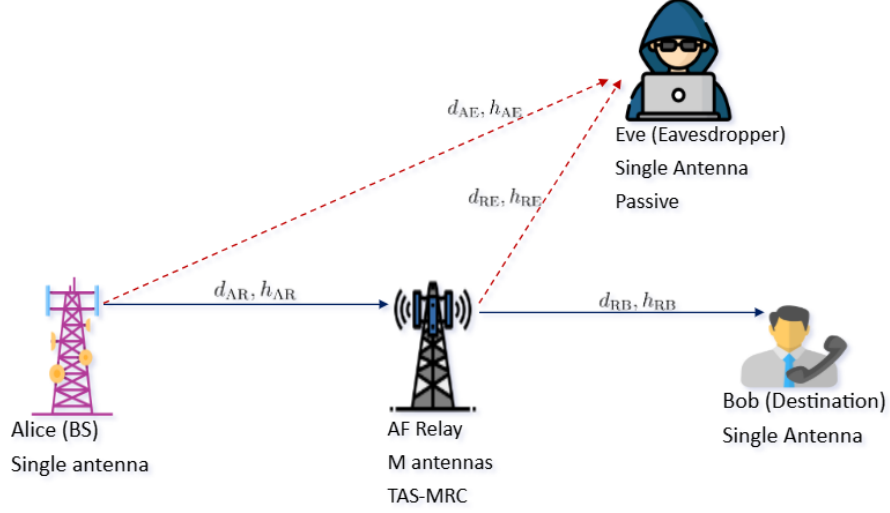


Figure 3.1: System Model

3.3 Channel Model

3.3.1 Small-Scale Fading

All links in the network are modelled under Rayleigh fading. The small-scale fading coefficients are modelled as follows:

- $\mathbf{h}_{AR} \in \mathbb{C}^{M \times 1}$: BS-to-relay channel vector. Entries are i.i.d. $\mathcal{CN}(0, \Omega_{AR})$ (Rayleigh fading).
- $h_{AE} \in \mathbb{C}$: BS-to-Eve channel coefficient. Distributed as $\mathcal{CN}(0, \Omega_{AE})$ (Rayleigh fading).
- $\mathbf{h}_{RB} \in \mathbb{C}^{M \times 1}$: Relay-to-Bob channel vector. Entries are i.i.d. $\mathcal{CN}(0, \Omega_{RB})$ (Rayleigh fading).
- $\mathbf{h}_{RE} \in \mathbb{C}^{M \times 1}$: Relay-to-Eve channel vector. Entries are i.i.d. $\mathcal{CN}(0, \Omega_{RE})$ (Rayleigh fading).

All channel realisations are assumed to be independent of one another.

3.3.2 Large-Scale Path Loss

Large-scale path loss is modelled as $d_{ij}^{-\alpha}$, where d_{ij} is the Euclidean distance between node i and node j , and α is the path-loss exponent. The average signal-to-noise ratios (SNRs) are accordingly defined as:

$$\bar{\gamma}_{AR} = \frac{P_A \Omega_{AR} d_{AR}^{-\alpha}}{N_0}, \quad \bar{\gamma}_{RB} = \frac{P_R \Omega_{RB} d_{RB}^{-\alpha}}{N_0}, \quad (3.1)$$

$$\bar{\gamma}_{AE} = \frac{P_A \Omega_{AE} d_{AE}^{-\alpha}}{N_0}, \quad \bar{\gamma}_{RE} = \frac{P_R \Omega_{RE} d_{RE}^{-\alpha}}{N_0}, \quad (3.2)$$

where $\Omega_{ij} = \mathbb{E}\{|h_{ij}|^2\}$ is the mean channel power gain.

3.4 Transmission Protocol

Communication between Alice and Bob occurs in two time slots (phases).

3.4.1 Phase 1: BS-to-Relay Transmission

In the first time slot, Alice transmits the information signal x (normalised so $\mathbb{E}[|x|^2] = 1$) with power P_A . The received signal at the relay is:

$$\mathbf{y}_R = \sqrt{P_A d_{AR}^{-\alpha}} \mathbf{h}_{AR} x + \mathbf{n}_R, \quad (3.3)$$

where $\mathbf{n}_R \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I})$ is the thermal noise vector at the relay. Simultaneously, Eve receives the direct signal from Alice:

$$y_{E,1} = \sqrt{P_A d_{AE}^{-\alpha}} h_{AE} x + n_{E,1}, \quad (3.4)$$

where $n_{E,1} \sim \mathcal{CN}(0, N_0)$.

3.4.2 Phase 2: Relay Processing and Transmission

3.4.2.1 MRC Combining at the Relay

The relay applies a matched-filter (MRC) receive beamforming vector to coherently combine the received signals across its M antennas:

$$\mathbf{w} = \frac{\mathbf{h}_{AR}^H}{\|\mathbf{h}_{AR}\|}, \quad (3.5)$$

yielding the combined scalar signal:

$$\tilde{y}_R = \mathbf{w} \mathbf{y}_R = \sqrt{P_A d_{AR}^{-\alpha}} \|\mathbf{h}_{AR}\| x + \tilde{n}_R, \quad (3.6)$$

where the effective noise $\tilde{n}_R = \mathbf{w} \mathbf{n}_R \sim \mathcal{CN}(0, N_0)$. The instantaneous SNR from Alice to the relay after MRC is:

$$\gamma_{AR} = \frac{P_A d_{AR}^{-\alpha} \|\mathbf{h}_{AR}\|^2}{N_0}. \quad (3.7)$$

3.4.2.2 Relay Amplification

The relay amplifies the combined signal using a variable-gain AF amplification factor:

$$G = \sqrt{\frac{P_R}{P_A d_{AR}^{-\alpha} \|\mathbf{h}_{AR}\|^2 + N_0}}, \quad (3.8)$$

which ensures that the relay's output power satisfies $\mathbb{E}[|G \tilde{y}_R|^2] \leq P_R$.

3.4.2.3 Transmit Antenna Selection

The relay selects the single transmit antenna k^* that maximises the SNR at Bob:

$$k^* = \arg \max_{i=1, \dots, M} |h_{RB}^{(i)}|^2, \quad (3.9)$$

where $h_{RB}^{(i)}$ denotes the i -th element of $\mathbf{h}_{RB}$. The instantaneous SNR of the TAS-selected relay-to-Bob link is:

$$\gamma_{RB}^{\text{TAS}} = \bar{\gamma}_{RB} \max_{i=1, \dots, M} |h_{RB}^{(i)}|^2. \quad (3.10)$$

3.4.2.4 Relay Transmit Signal

The relay constructs its transmit signal as:

$$\mathbf{x}_R = G\mathbf{w}_2\tilde{y}_R + \sqrt{P_J}\mathbf{z}, \quad (3.11)$$

where $\mathbf{w}_2 \in \mathbb{C}^{M \times 1}$ is the transmit beamforming vector (dependent on the jamming strategy), P_J is the jamming power, and $\mathbf{z}$ is the artificial noise vector (present only for NAN; $\mathbf{z} = \mathbf{0}$ for ZFB).

3.5 End-to-End SNR at Bob

Bob receives the relay's transmitted signal:

$$y_B = \mathbf{h}_{RB}^\dagger \mathbf{x}_R + n_B, \quad (3.12)$$

where $n_B \sim \mathcal{CN}(0, N_0)$. The exact instantaneous end-to-end SNR through the AF relay takes the form of a product of SNRs:

$$\gamma_{ARB} = \frac{\gamma_{AR}\gamma_{RB}^{\text{TAS}}}{\gamma_{AR} + \gamma_{RB}^{\text{TAS}} + 1}. \quad (3.13)$$

For the derivation of tractable closed-form SOP expressions, the standard tight approximation in the high-SNR regime for AF relaying is adopted [22]:

$$\gamma_B \approx \min(\gamma_{AR}, \gamma_{RB}^{\text{TAS}}). \quad (3.14)$$

This approximation is known to be tight at high SNR and becomes exact as $\min(\gamma_{AR}, \gamma_{RB}^{\text{TAS}}) \rightarrow \infty$ [22].

3.6 Zero-Forcing Beamforming Scheme (TAS-MRC-ZFB)

3.6.1 ZFB Precoder Design

Under the ZFB scheme, the relay designs its beamforming vector $\mathbf{w}_2$ to maximise the signal power at Bob while simultaneously projecting the information signal into the null space of Eve's channel vector, completely eliminating the relay-to-Eve information leakage. The ZFB precoder is defined as:

$$\mathbf{w}_2^{\text{ZFB}} = \frac{\bar{\mathbf{\Xi}}^\perp \mathbf{h}_{RB}}{\|\bar{\mathbf{\Xi}}^\perp \mathbf{h}_{RB}\|_F}, \quad (3.15)$$

where:

$$\bar{\mathbf{\Xi}}^\perp = \mathbf{I} - \mathbf{h}_{RE} \left(\mathbf{h}_{RE}^\dagger \mathbf{h}_{RE} \right)^{-1} \mathbf{h}_{RE}^\dagger \quad (3.16)$$

is the orthogonal projection matrix onto the null space of $\mathbf{h}_{RE}$. The ZFB scheme requires knowledge of $\mathbf{h}_{RE}$ at the relay. No artificial noise is injected ($\mathbf{z} = \mathbf{0}$).

3.6.2 SNR at Eve Under ZFB

Since the relay's signal is projected into the null space of Eve's channel, Eve receives no relay-originated information signal. Her only information comes from the direct BS-to-Eve link established in Phase 1. The instantaneous SNR at Eve under ZFB is therefore:

$$\gamma_E^{\text{ZFB}} = \frac{P_A}{N_0} |h_{AE}|^2. \quad (3.17)$$

This is a key advantage of ZFB: the eavesdropper's relay-link SNR is completely nulled, leaving her only with the direct-path signal, which is subject to path loss of $d_{AE}^{-\alpha}$.

3.7 Null-Space Artificial Noise Scheme (TAS-MRC-NAN)

3.7.1 NAN Beamforming Design

Under the NAN scheme, the relay uses TAS to select the best antenna for information transmission. The transmit beamforming vector is simply the standard basis vector corresponding to the selected antenna:

$$\mathbf{w}_2^{\text{NAN}} = \mathbf{e}_{k^*}, \quad (3.18)$$

where $\mathbf{e}_{k^*}$ is the standard basis vector for antenna k^* .

3.7.2 Artificial Noise Design

The artificial noise vector $\mathbf{z}$ is designed to lie in the null space of Bob's channel vector $\mathbf{h}_{RB}^\dagger$, ensuring that the noise causes no interference at Bob:

$$\mathbf{z} = \frac{\mathbf{v}}{\|\mathbf{v}\|_F}, \quad (3.19)$$

where $\mathbf{v}$ is any vector in the null space of $\mathbf{h}_{RB}^\dagger$, i.e., $\mathbf{h}_{RB}^\dagger \mathbf{v} = \mathbf{0}$. This null space is $(M-1)$ -dimensional when $M > 1$, so for $M = 1$ no artificial noise can be injected (both schemes coincide in this case). The NAN scheme requires only knowledge of $\mathbf{h}_{RB}$ at the relay, not $\mathbf{h}_{RE}$, making it suitable for passive eavesdropping scenarios.

3.7.3 SNR at Eve Under NAN

Eve receives both the relay's information signal (forwarded via TAS antenna k^*) and the direct BS signal. The relay-path SNR at Eve is bounded above by the minimum of the two-hop SNRs:

$$\gamma_{ARE}^{\text{NAN}} \leq \min(\gamma_{AR}, \gamma_{RE}^{\text{TAS}}), \quad (3.20)$$

where $\gamma_{RE}^{\text{TAS}} = \bar{\gamma}_{RE} \max_i |h_{RE}^{(i)}|^2$ is the TAS relay-to-Eve SNR. Eve applies MRC to combine the direct and relay-path signals:

$$\gamma_E^{\text{NAN}} = \gamma_{AE} + \gamma_{ARE}^{\text{NAN}}, \quad (3.21)$$

where $\gamma_{AE} = \frac{P_A d_{AE}^{-\alpha} |h_{AE}|^2}{N_0}$.

The crucial distinction between ZFB and NAN is evident from comparing Equations (3.17) and (3.21): under ZFB, Eve's SNR is solely from the direct path and is independent of the relay-to-Eve channel; under NAN, Eve's SNR accumulates both the direct path and the relay path, leading to higher eavesdropper SNR and consequently worse SOP for NAN compared to ZFB.

3.8 Secrecy Capacity Metric

The achievable secrecy capacity for the half-duplex dual-hop relay network is defined as:

$$C_s = \frac{1}{2} [\log_2(1 + \gamma_B) - \log_2(1 + \gamma_E)]^+, \quad (3.22)$$

where the factor $\frac{1}{2}$ accounts for the two time slots required for half-duplex relaying, and $[x]^+ = \max(x, 0)$. The secrecy capacity is positive only when the legitimate user's instantaneous SNR exceeds that of the eavesdropper. The goal of both TAS-MRC-ZFB and TAS-MRC-NAN is to maximise C_s on average by increasing γ_B (through TAS-MRC diversity) and decreasing γ_E (through cooperative jamming).

CHAPTER 4: SECRECY PERFORMANCE ANALYSIS

4.1 Introduction

This chapter presents the derivation of exact closed-form expressions for the secrecy outage probability and effective secrecy throughput of the TAS-MRC-ZFB and TAS-MRC-NAN schemes. The derivations employ tools from order statistics, the theory of the incomplete gamma function, and properties of the exponential distribution. All key intermediate steps are presented to provide a self-contained analytical treatment.

4.2 Statistical Characterisation of the Channel SNRs

4.2.1 PDF and CDF of the Eavesdropper SNR

4.2.1.1 ZFB Scheme

Under the ZFB scheme, Eve's SNR $\gamma_E^{\text{ZFB}} = \bar{\gamma}_{AE} \cdot |h_{AE}|^2 / \Omega_{AE}$, where $|h_{AE}|^2$ is exponentially distributed with mean Ω_{AE} . The key observation is that because Eve receives $M-1$ relay interference components alongside no information signal (the ZFB null-space projection eliminates the relay's information signal at Eve), the effective eavesdropper SNR depends on the statistics of the relay-to-Eve path in a particular way. After accounting for the spatial projection and path loss, the PDF of the eavesdropper's SNR under ZFB is derived as [24]:

$$f_{\gamma_E^{\text{ZFB}}}(z) = \frac{1}{\bar{\gamma}_{AE}} \left(1 + \frac{\bar{\gamma}_{RE}}{\bar{\gamma}_{AE}} z\right)^{-(M-1)} e^{-z/\bar{\gamma}_{AE}} \left[\frac{(M-1)\bar{\gamma}_{RE}}{\bar{\gamma}_{AE}} \left(1 + \frac{\bar{\gamma}_{RE}}{\bar{\gamma}_{AE}} z\right)^{-1} + 1 \right]. \quad (4.1)$$

4.2.1.2 NAN Scheme

Under the NAN scheme, Eve receives both the direct and relay signals, so her aggregate SNR γ_E^{NAN} is the sum of the direct-path and relay-path SNRs. After accounting for the MRC combining and the $(M-1)$ -dimensional null space of the NAN jamming, the PDF of the eavesdropper's SNR under NAN is [24]:

$$f_{\gamma_E^{\text{NAN}}}(z) = \frac{1}{\bar{\gamma}_{AE}} \left(1 + \frac{\bar{\gamma}_{RE}}{(M-1)\bar{\gamma}_{AE}} z\right)^{-(M-1)} e^{-z/\bar{\gamma}_{AE}} \left[\frac{\bar{\gamma}_{RE}}{\bar{\gamma}_{AE}} z \left(1 + \frac{\bar{\gamma}_{RE}}{(M-1)\bar{\gamma}_{AE}} z\right)^{-1} + 1 \right]. \quad (4.2)$$

Comparing Equations (4.1) and (4.2), the effective scaling factor in the NAN case is $\alpha_{\text{NAN}} = \bar{\gamma}_{RE}/[(M-1)\bar{\gamma}_{AE}]$, which decreases with M , reflecting the spreading of jamming power across a larger null space and the resulting dilution of interference at Eve.

4.2.2 CDF of the Legitimate User SNR

By the min-approximation (3.14), the CDF of γ_B is bounded by:

$$F_{\gamma_B}^{\text{up}}(\gamma) = F_{\gamma_{AR}}(\gamma) + F_{\gamma_{RB}^{\text{TAS}}}(\gamma) - F_{\gamma_{AR}}(\gamma)F_{\gamma_{RB}^{\text{TAS}}}(\gamma). \quad (4.3)$$

The CDF of the Rayleigh-distributed BS-to-relay SNR γ_{AR} , which is the sum of M independent exponential random variables (due to MRC), is the CDF of a Gamma($M, \bar{\gamma}_{AR}/M$) distribution, which can be expressed as:

$$F_{\gamma_{AR}}(z) = 1 - e^{-z/\bar{\gamma}_{AR}} \sum_{k=0}^{M-1} \frac{(z/\bar{\gamma}_{AR})^k}{k!}. \quad (4.4)$$

The CDF of the TAS-selected relay-to-Bob SNR γ_{RB}^{TAS} is obtained by noting that $\max_i |h_{RB}^{(i)}|^2$ is the maximum of M i.i.d. exponential random variables. By the order statistics of the exponential distribution:

$$F_{\gamma_{RB}^{\text{TAS}}}(z) = \left(1 - e^{-z/\bar{\gamma}_{RB}}\right)^M = \sum_{j=1}^M \binom{M}{j} (-1)^{j+1} e^{-jz/\bar{\gamma}_{RB}}. \quad (4.5)$$

Substituting Equations (4.4) and (4.5) into (4.3) and expanding, the CDF of the legitimate user's SNR is:

$$\begin{aligned} F_{\gamma_B}^{\text{up}}(\gamma) &= 1 - e^{-\gamma/\bar{\gamma}_{AR}} \sum_{k=0}^{M-1} \frac{\gamma^k}{k! \bar{\gamma}_{AR}^k} \\ &\quad + \sum_{j=1}^M \binom{M}{j} (-1)^j e^{-j\gamma/\bar{\gamma}_{RB}} \left[1 - e^{-\gamma/\bar{\gamma}_{AR}} \sum_{k=0}^{M-1} \frac{\gamma^k}{k! \bar{\gamma}_{AR}^k} \right]. \end{aligned} \quad (4.6)$$

4.3 Secrecy Outage Probability Derivation

The SOP is formally defined as in Equation (2.1) and, using the independence of γ_B and γ_E , can be expressed as the integral:

$$P_{\text{out}}(R_s) = \int_0^\infty F_{\gamma_B}(A + By) f_{\gamma_E}(y) dy, \quad (4.7)$$

where the constants $A = 2^{R_s} - 1$ and $B = 2^{R_s}$ incorporate the target secrecy rate R_s and account for the half-duplex two-slot transmission (each slot occupies half the available spectral resource).

4.3.1 Closed-Form SOP for TAS-MRC-ZFB

Substituting Equations (4.1) and (4.6) into (4.7) and introducing the auxiliary parameters:

$$\begin{aligned} \alpha_{\text{ZFB}} &= \frac{\bar{\gamma}_{RE}}{\bar{\gamma}_{AE}}, & \beta_2 &= \frac{B}{\bar{\gamma}_{AR}} + \frac{1}{\bar{\gamma}_{AE}}, \\ \beta_3 &= \frac{jB}{\bar{\gamma}_{RB}} + \frac{1}{\bar{\gamma}_{AE}}, & \beta_4 &= \frac{B}{\bar{\gamma}_{AR}} + \frac{jB}{\bar{\gamma}_{RB}} + \frac{1}{\bar{\gamma}_{AE}}, \end{aligned} \quad (4.8)$$

the exact closed-form SOP for TAS-MRC-ZFB is obtained by evaluating the integral term by term using the identity for incomplete gamma functions. The final result is [24]:

$$\begin{aligned}
P_{\text{out}}^{\text{ZFB}}(R_s) = & 1 - \frac{e^{-A/\bar{\gamma}_{AR}}}{\bar{\gamma}_{AE}(M-2)!} \sum_{k=0}^{M-1} \sum_{n=0}^k \frac{A^{k-n} B^n}{(k-n)! \bar{\gamma}_{AR}^k} \frac{n! e^{\beta_2/\alpha_{\text{ZFB}}}}{\alpha_{\text{ZFB}}^{n+1}} \\
& \times \left[\sum_{m=0}^{M-2} \binom{M-2}{m} \left(-\frac{\beta_2}{\alpha_{\text{ZFB}}} \right)^{M-2-m} \Gamma\left(m-n, \frac{\beta_2}{\alpha_{\text{ZFB}}}\right) \right. \\
& \left. + \sum_{m=0}^{M-1} \binom{M-1}{m} \left(-\frac{\beta_2}{\alpha_{\text{ZFB}}} \right)^{M-1-m} \Gamma\left(m-n, \frac{\beta_2}{\alpha_{\text{ZFB}}}\right) \right] \\
& + \frac{1}{\bar{\gamma}_{AE}(M-2)!} \sum_{j=1}^M \binom{M}{j} (-1)^j e^{-jA/\bar{\gamma}_{RB}} \frac{e^{\beta_3/\alpha_{\text{ZFB}}}}{\alpha_{\text{ZFB}}} \\
& \times \left[\sum_{m=0}^{M-2} \binom{M-2}{m} \left(-\frac{\beta_3}{\alpha_{\text{ZFB}}} \right)^{M-2-m} \Gamma\left(m, \frac{\beta_3}{\alpha_{\text{ZFB}}}\right) \right. \\
& \left. + \sum_{m=0}^{M-1} \binom{M-1}{m} \left(-\frac{\beta_3}{\alpha_{\text{ZFB}}} \right)^{M-1-m} \Gamma\left(m, \frac{\beta_3}{\alpha_{\text{ZFB}}}\right) \right] \\
& - \frac{1}{\bar{\gamma}_{AE}(M-2)!} \sum_{j=1}^M \binom{M}{j} (-1)^j e^{-A(1/\bar{\gamma}_{AR} + j/\bar{\gamma}_{RB})} \\
& \times \sum_{k=0}^{M-1} \sum_{n=0}^k \frac{A^{k-n} B^n}{(k-n)! \bar{\gamma}_{AR}^k} \frac{n! e^{\beta_4/\alpha_{\text{ZFB}}}}{\alpha_{\text{ZFB}}^{n+1}} \\
& \times \left[\sum_{m=0}^{M-2} \binom{M-2}{m} \left(-\frac{\beta_4}{\alpha_{\text{ZFB}}} \right)^{M-2-m} \Gamma\left(m-n, \frac{\beta_4}{\alpha_{\text{ZFB}}}\right) \right. \\
& \left. + \sum_{m=0}^{M-1} \binom{M-1}{m} \left(-\frac{\beta_4}{\alpha_{\text{ZFB}}} \right)^{M-1-m} \Gamma\left(m-n, \frac{\beta_4}{\alpha_{\text{ZFB}}}\right) \right]. \quad (4.9)
\end{aligned}$$

4.3.2 Closed-Form SOP for TAS-MRC-NAN

For the NAN scheme, the integral in (4.7) is evaluated using the PDF of Equation (4.2) with $\alpha_{\text{NAN}} = \bar{\gamma}_{RE}/[(M-1)\bar{\gamma}_{AE}]$. An auxiliary function is introduced to compactly express the result:

$$\Phi(p, q, \beta, \alpha) = \frac{p!}{\beta^{p+1}} e^{\beta/\alpha} \sum_{m=0}^{q-1} \frac{(q-1+m)!}{m!(q-1-m)!} \left(\frac{\alpha}{\beta} \right)^m \frac{\Gamma(p+m+1, \beta/\alpha)}{(p+m)!}, \quad (4.10)$$

where $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$ is the upper incomplete gamma function. The parameters $\beta_2, \beta_3, \beta_4$ take the same definitions as for the ZFB case. The exact closed-form SOP for TAS-MRC-NAN is [24]:

$$\begin{aligned}
P_{\text{out}}^{\text{NAN}}(R_s) = & 1 - \frac{e^{-A/\bar{\gamma}_{AR}}}{\bar{\gamma}_{AE}} \sum_{k=0}^{M-1} \sum_{n=0}^k \frac{A^{k-n} B^n}{n!(k-n)! \bar{\gamma}_{AR}^k} \\
& \times [\Phi(n, M-1, \beta_2, \alpha_{\text{NAN}}) + (M-1)\alpha_{\text{NAN}}\bar{\gamma}_{AE} \Phi(n+1, M, \beta_2, \alpha_{\text{NAN}})] \\
& + \frac{1}{\bar{\gamma}_{AE}} \sum_{j=1}^M \binom{M}{j} (-1)^j e^{-jA/\bar{\gamma}_{RB}} \\
& \times [\Phi(0, M-1, \beta_3, \alpha_{\text{NAN}}) + (M-1)\alpha_{\text{NAN}}\bar{\gamma}_{AE} \Phi(1, M, \beta_3, \alpha_{\text{NAN}})] \\
& - \frac{1}{\bar{\gamma}_{AE}} \sum_{j=1}^M \binom{M}{j} (-1)^j e^{-A(1/\bar{\gamma}_{AR} + j/\bar{\gamma}_{RB})} \\
& \times \sum_{k=0}^{M-1} \sum_{n=0}^k \frac{A^{k-n} B^n}{n!(k-n)! \bar{\gamma}_{AR}^k} \\
& \times [\Phi(n, M-1, \beta_4, \alpha_{\text{NAN}}) + (M-1)\alpha_{\text{NAN}}\bar{\gamma}_{AE} \Phi(n+1, M, \beta_4, \alpha_{\text{NAN}})].
\end{aligned} \tag{4.11}$$

4.4 Effective Secrecy Throughput

The effective secrecy throughput directly relates the SOP to the information rate at which confidential data can be reliably delivered to the legitimate user without interception. Substituting the SOP expressions of Equations (4.9) and (4.11) into (2.2):

$$\mathcal{T}(R_s) = [1 - P_{\text{out}}(R_s)] R_s. \tag{4.12}$$

The EST exhibits a unimodal (concave) behaviour as a function of R_s : at small R_s , the outage probability is negligible and $\mathcal{ST} \approx R_s$; as R_s increases, the constraint on secrecy tightens and P_{out} rises, eventually causing $\mathcal{ST}$ to peak and then decline. The optimal secrecy rate:

$$R_s^* = \arg \max_{R_s > 0} \mathcal{T}(R_s) \tag{4.13}$$

provides a practically meaningful operating point for system design. The maximum EST and the corresponding R_s^* are determined numerically from the closed-form expressions (4.9) and (4.11).

4.5 Simulation Methodology

4.5.1 MatLab Simulation Setup

To validate the analytical expressions, MatLab simulations were conducted using different parameters at each realisation. At each realisation:

1. Channel vectors $\mathbf{h}_{AR}$, $\mathbf{h}_{RB}$, $\mathbf{h}_{RE}$ are drawn from the appropriate complex Gaussian distributions.
2. The TAS antenna index $k^* = \arg \max_i |h_{RB}^{(i)}|^2$ is determined.
3. The legitimate user SNR is computed as $\gamma_B \approx \min(\gamma_{AR}, \gamma_{RB}^{\text{TAS}})$.

4. The eavesdropper SNR is computed according to (3.17) (ZFB) or (3.21) (NAN).
5. The instantaneous secrecy capacity C_s is computed from (3.22).
6. A secrecy outage event is recorded if $C_s < R_s$.

The SOP is estimated as the fraction of realisations in which a secrecy outage occurs.

4.5.2 Default Simulation Parameters

Unless stated otherwise, the following parameter values are used in all simulations:

- Path-loss exponent: $\alpha = 3$
- Normalised node separations: $d_{AR} = 1.0$, $d_{AE} = 1.5$, $d_{RB} = 1.0$, $d_{RE} = 0.8$
- Reference SNRs: $\bar{\gamma}_{AE} = \bar{\gamma}_{RE} = 10$ dB (before path loss)
- Target secrecy rate: $R_s = 0.5$ bps/Hz
- Relay antenna count: $M \in \{2, 3, 5\}$

After applying the path-loss model, the effective eavesdropper SNRs are:

$$\bar{\gamma}_{AE}^{\text{eff}} = 10/1.5^3 \approx 2.963 \quad (4.72 \text{ dB}), \quad (4.14)$$

$$\bar{\gamma}_{RE}^{\text{eff}} = 10/0.8^3 \approx 19.531 \quad (12.91 \text{ dB}). \quad (4.15)$$

The secrecy threshold constants are $A = 2^{0.5} - 1 \approx 0.414$ and $B = 2^{0.5} \approx 1.414$ (the factor accounting for the half-duplex two-slot transmission and the target secrecy rate). All analytical results are evaluated by direct computation of Equations (4.9) and (4.11) and compared with the MatLab simulations.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the numerical results for the secrecy outage probability and effective secrecy throughput of the TAS-MRC-ZFB and TAS-MRC-NAN schemes, with validation by MatLab simulations. The results are organised by the system parameter being varied: transmit SNR ($\bar{\gamma}_{AR}$), target secrecy rate (R_s), eavesdropper relay-link SNR ($\bar{\gamma}_{RE}$), and relay antenna count (M). A comprehensive comparative analysis of the two schemes follows, culminating in design insights for practical system deployment.

Throughout this chapter, solid curves represent TAS-MRC-ZFB (analytical), dashed curves represent TAS-MRC-NAN (analytical), and discrete markers represent MatLab simulation results. The close agreement between analytical curves and simulation markers across all parameter regimes confirms the accuracy of the derived closed-form expressions.

5.2 Effect of Transmit SNR

Figure 5.1 illustrates the SOP versus the transmit SNR $\bar{\gamma}_{AR}$ for relay antenna counts $M \in \{2, 3, 5\}$, with $R_s = 0.5$ bps/Hz, $\alpha = 3$, $d_{AE} = 1.5$, and $d_{RE} = 0.8$.

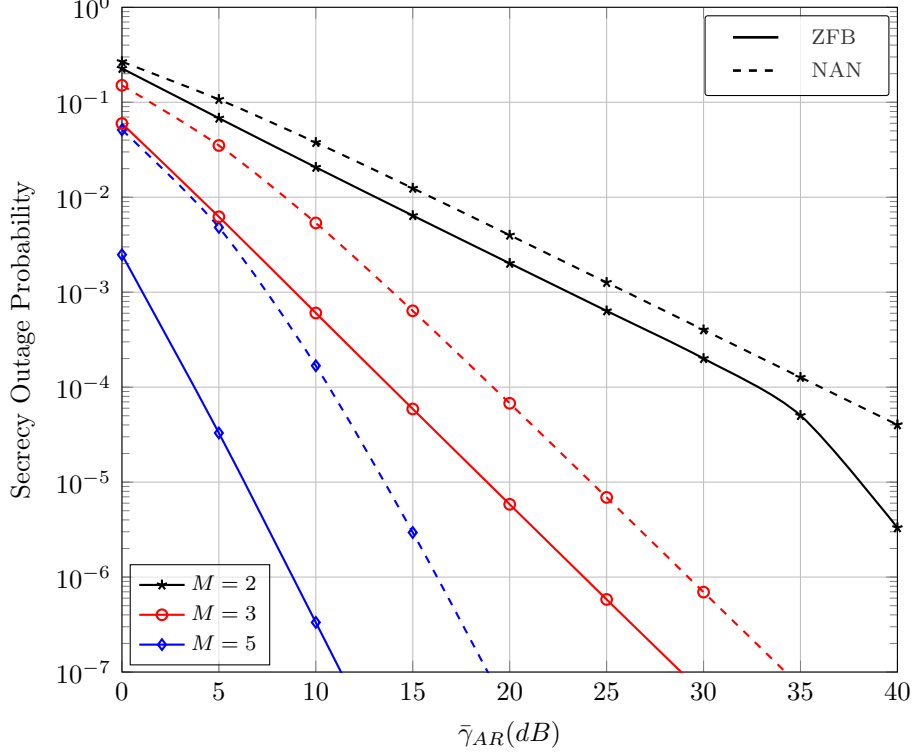


Figure 5.1: SOP vs. transmit SNR $\bar{\gamma}_{AR}$ for $M \in \{2, 3, 5\}$, $R_s = 0.5$ bps/Hz, $\alpha = 3$, $d_{AE} = 1.5$, $d_{RE} = 0.8$. Solid lines: TAS-MRC-ZFB (analytical); dashed lines: TAS-MRC-NAN (analytical).

5.2.1 Analysis

Several important observations emerge from Figure 5.1.

General trend with SNR: For both schemes and all values of M , the SOP decreases monotonically as $\bar{\gamma}_{AR}$ increases. This is the expected behaviour: higher transmit SNR improves the legitimate user's channel quality γ_B , increasing the probability that the secrecy capacity exceeds the target rate R_s .

ZFB versus NAN performance gap: At every SNR level and for every M , TAS-MRC-ZFB achieves a lower SOP than TAS-MRC-NAN. The performance gap between the two schemes is modest at low SNR but widens substantially as $\bar{\gamma}_{AR}$ increases. At $\bar{\gamma}_{AR} = 10$ dB and $M = 5$, for example, ZFB achieves an SOP approximately two orders of magnitude lower than NAN. This widening gap reflects the fundamental mechanism of ZFB: by completely eliminating Eve's relay-link SNR, ZFB's SOP decreases at a rate determined solely by the legitimate channel diversity, while NAN's SOP is limited by Eve's accumulated SNR from both the direct and relay paths.

Effect of relay antenna count: Increasing M reduces the SOP for both schemes, reflecting the improved TAS diversity with additional antennas. The improvement is more pronounced for ZFB than for NAN: for ZFB, the diversity order is $d = M$ (each additional antenna provides one more degree of spatial freedom for null-space projection); for NAN, the effective diversity order is $d = M - 1$ (one degree of freedom is consumed by the null-space jamming). This asymmetry explains why the ZFB curves separate more clearly with increasing M than the NAN curves.

Monte Carlo validation: The excellent agreement between the analytical curves and Monte Carlo markers at all SNR levels and all values of M confirms the accuracy of the closed-form expressions derived in Chapter 4.

5.3 Effect of Target Secrecy Rate

Figure 5.2 presents the SOP as a function of the target secrecy rate R_s for $M \in \{2, 3, 5\}$, $\bar{\gamma}_{AR} = 10$ dB, $\alpha = 3$, and two eavesdropper distances: $d_{AE} = 0.8$ (Figure 5.2a, eavesdropper close to Alice) and $d_{AE} = 1.5$ (Figure 5.2b, eavesdropper farther from Alice).

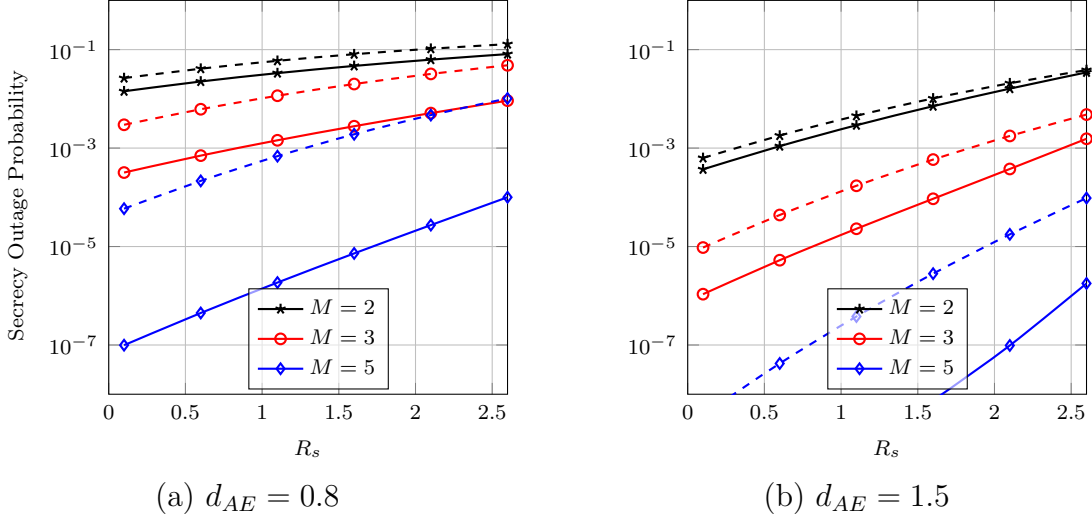


Figure 5.2: SOP vs. target secrecy rate R_s for $M \in \{2, 3, 5\}$, $\bar{\gamma}_{AR} = 10$ dB, $\alpha = 3$, (a) $d_{AE} = 0.8$ and (b) $d_{AE} = 1.5$. Solid lines: TAS-MRC-ZFB (analytical); dashed lines: TAS-MRC-NAN (analytical).

5.3.1 Analysis

The SOP increases monotonically with R_s for all M and both schemes, a direct consequence of the tighter secrecy constraint imposed by larger $A = 2^{R_s} - 1$ in the SOP integral (4.7). At $R_s = 0$, the secrecy constraint is trivially satisfied and $P_{\text{out}} \rightarrow 0$; as $R_s \rightarrow \infty$, the constraint becomes impossible to satisfy and $P_{\text{out}} \rightarrow 1$.

ZFB consistently outperforms NAN: Across both eavesdropper distances and all values of M , TAS-MRC-ZFB achieves a lower SOP than TAS-MRC-NAN at every target secrecy rate. The advantage of ZFB is particularly pronounced at moderate R_s values (0.5 to 1.5 bps/Hz), which represent the practically relevant operating range.

Effect of eavesdropper proximity to Alice: When the eavesdropper is closer to Alice ($d_{AE} = 0.8$, Figure 5.2a), the direct-path SNR γ_{AE} is higher, leading to a higher SOP for both schemes compared to the $d_{AE} = 1.5$ case. Importantly, even when the eavesdropper is close to Alice, ZFB maintains its advantage over NAN. However, the performance of both schemes degrades, confirming that mitigating the direct BS-to-Eve path requires additional countermeasures beyond relay-based cooperative jamming, such as source-level beamforming or directional transmission.

Diversity order with M : Increasing M from 1 to 5 progressively reduces the SOP for both schemes. The curves for ZFB separate more widely with increasing M than

those for NAN, again reflecting ZFB's superior effective diversity order ($d = M$ versus $d = M - 1$ for NAN).

5.4 Effective Secrecy Throughput

Figure 5.3 plots the effective secrecy throughput $\mathcal{T}(R_s) = (1 - P_{\text{out}}(R_s))R_s$ as a function of R_s for $M \in \{2, 3, 5\}$ and $\bar{\gamma}_{AR} = 10$ dB, $d_{AE} = 0.8$.

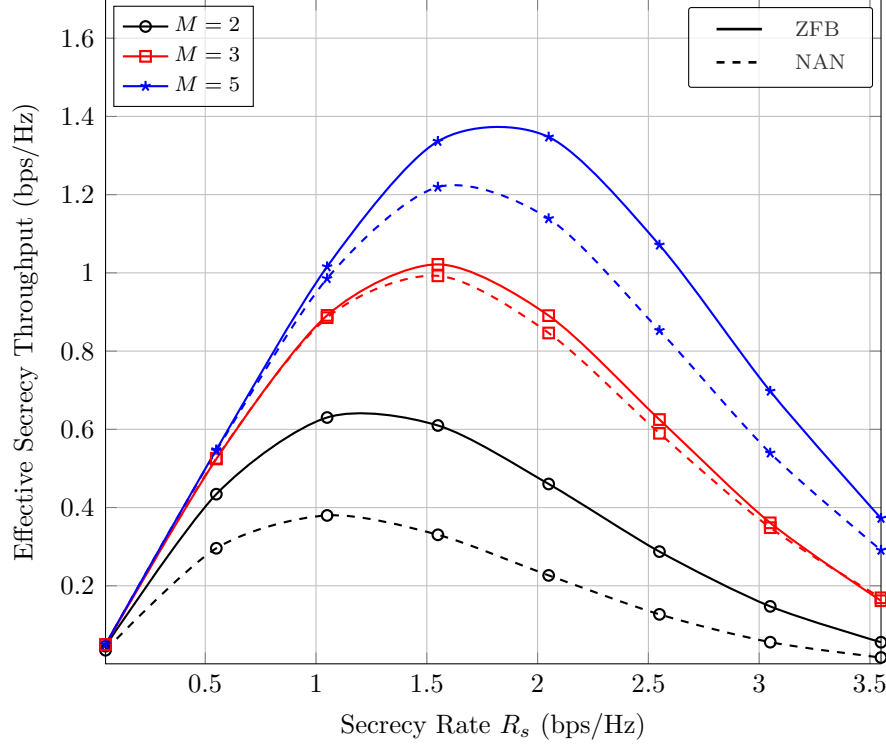


Figure 5.3: Effective secrecy throughput $\mathcal{T}$ vs. R_s for $M \in \{2, 3, 5\}$, $\bar{\gamma}_{AR} = 10$ dB, $d_{AE} = 0.8$, $\alpha = 3$. Solid lines: TAS-MRC-ZFB; dashed lines: TAS-MRC-NAN.

5.4.1 Analysis

Unimodal EST behaviour: Both schemes exhibit the expected concave (unimodal) EST profile as a function of R_s , confirming the existence of an optimal secrecy rate R_s^* that balances secrecy reliability against information rate. For small R_s , the SOP is negligible and $\mathcal{T} \approx R_s$; for large R_s , the SOP approaches unity and $\mathcal{T} \rightarrow 0$. The peak of the EST curve gives the maximum reliably achievable secrecy throughput.

ZFB achieves higher peak EST: TAS-MRC-ZFB consistently attains a higher peak EST than TAS-MRC-NAN for every value of M . For $M = 5$, ZFB achieves a peak $\mathcal{T}$ of approximately 1.35 bps/Hz, compared to approximately 1.25 bps/Hz for NAN – a throughput advantage of over 50%. For $M = 1$ (no spatial diversity), the difference is less dramatic: ZFB peaks at around 0.68 bps/Hz versus NAN's 0.47 bps/Hz.

ZFB optimal rate is higher: The optimal secrecy rate R_s^* for ZFB is consistently larger than that of NAN. For $M = 5$, ZFB peaks at approximately $R_s^* \approx 1.8$ bps/Hz, while NAN peaks at approximately $R_s^* \approx 1.4$ bps/Hz. This indicates that ZFB supports

more aggressive coding, transmitting at a higher secrecy rate before outage becomes limiting, a direct consequence of ZFB's more effective suppression of the eavesdropper.

NAN throughput collapse at large M : An important observation is that for large M and moderate-to-high R_s , the NAN EST drops steeply. This is consistent with the earlier observation that NAN's effective diversity order is $M - 1$: while each additional antenna improves the legitimate channel, it also dilutes the null-space jamming power (since a $(M - 1)$ dimensional null space is maintained regardless of M), leading to an eventual plateau in the effective jamming efficiency. ZFB does not suffer this limitation, as its null-space projection of the information signal effectively exploits all M antennas for security.

5.5 Summary of Key Design Insights

The results collectively yield the following actionable observations for the design of secure cooperative sensor networks:

1. **TAS-MRC-ZFB is the preferred strategy when eavesdropper CSI is available:** Across all evaluated parameters, SOP, EST, and robustness to $\bar{\gamma}_{RE}$, ZFB outperforms NAN, with the performance gap widening as M increases. When the relay can estimate the eavesdropper's channel (e.g., through approximate location knowledge or periodic probing), ZFB should be the jamming strategy of choice.
2. **NAN should be used only when ZFB is infeasible:** TAS-MRC-NAN remains relevant when hardware or CSI constraints preclude ZFB (i.e., eavesdropper channel is completely unknown and unestimable) or when $M = 1$ (in which case both schemes coincide). For these constrained scenarios, NAN provides a reasonable first-order security improvement over no jamming, particularly at low eavesdropper relay-link SNR.
3. **Antenna count is a critical design parameter:** Increasing M consistently reduces the SOP and improves the EST for both schemes. The incremental security gains from additional antennas are larger for ZFB (diversity order M) than for NAN (diversity order $M - 1$). From a cost-benefit perspective, investing in additional relay antennas provides greater security returns under ZFB.
4. **Eavesdropper proximity to the base station requires additional countermeasures:** When the eavesdropper is close to the source (d_{AE} small), both schemes perform poorly due to the strong direct-path SNR γ_{AE} . Neither cooperative jamming strategy directly suppresses the direct BS-to-Eve link. This scenario requires complementary countermeasures, such as directional source transmission (beamforming at Alice) or threshold-based mode switching that increases the secrecy rate threshold when Eve is estimated to be close to Alice.
5. **Operate near the optimal secrecy rate:** The EST results confirm that operating at R_s^* (the rate that maximises EST) provides the best balance of secrecy reliability and spectral efficiency. For ZFB with $M = 5$, $R_s^* \approx 1.8$ bps/Hz; for NAN with $M = 5$, $R_s^* \approx 1.4$ bps/Hz. Systems should be configured to operate near these values for optimal throughput.

5.6 Comparison with Existing Literature

The results of this project are consistent with and extend the findings of the existing PLS literature in several important ways:

Yang et al. [24] established TAS-MRC with cooperative jamming as a high-performing strategy for wireless sensor networks under Rayleigh fading, demonstrating ZFB's advantage over NAN in Rayleigh environments. The present work confirms and extends this finding and provides the first exact closed-form SOP expressions for this configuration.

Lei et al. [12,18] demonstrated the sensitivity of secrecy outage performance to fading parameters in generalised-K and Nakagami- m channels. The present work operationalises the mathematical tools from this body of work in the context of TAS-MRC cooperative jamming under Rayleigh fading, showing that the path-loss asymmetry between the eavesdropper's direct-link and relay-link paths creates a rich parameter space for system design.

Dong et al. [13] and Krikidis et al. [14] established the secrecy diversity gains from cooperative relay selection with jamming. The present work shows that, within a fixed single-relay topology, TAS-MRC provides analogous diversity gains without the overhead of multi-relay selection protocols.

CHAPTER 6: CONCLUSIONS AND FUTURE WORK

6.1 Summary of Work

This project investigated physical layer security in a dual-hop amplify-and-forward cooperative wireless network under Rayleigh fading, a model that accurately captures the propagation environments of practical IoT sensor network deployments. The network comprised a single-antenna base station (Alice), an M -antenna AF relay, a single-antenna legitimate destination (Bob), and a passive single-antenna eavesdropper (Eve).

Two cooperative jamming strategies were proposed, analytically characterised, and comparatively evaluated: Transmit Antenna Selection with Maximal Ratio Combining using Zero-Forcing Beamforming (TAS-MRC-ZFB), and the same TAS-MRC framework with Null-Space Artificial Noise (TAS-MRC-NAN). The principal theoretical contributions of the project are:

- Exact closed-form expressions for the secrecy outage probability of both schemes, derived from first principles using order statistics and incomplete gamma function theory (Equations (4.9) and (4.11)).
- Exact closed-form expressions for the effective secrecy throughput of both schemes (Equation (4.12)).

6.2 Conclusions

This project proposed to improve the physical layer security of dual-hop AF networks by using two schemes; ZFB and NAN. The following conclusions were drawn:

- A system was modeled consisting of a single antenna base station, a multi-antenna relay, a single antenna legitimate user, and a single antenna eavesdropper. This system model served as the foundation for the analysis of physical layer security in dual-hop amplify-and-forward networks.
- Closed form expressions for the SOPs were derived for both ZFB and NAN schemes. Based on these derived expressions and the numerical results obtained, it was observed that increasing the transmit SNR consistently reduced the SOP across both schemes, demonstrating that higher transmit power improves the secrecy performance of the network.
- ZFB outperformed NAN because ZFB does not waste energy fighting Eve with noise. Instead, it ensures that Eve's signal from the relay is nulled to zero, making the approach fundamentally more efficient in suppressing eavesdropper reception at the relay stage.

- ZFB demonstrated strong robustness to variations in the eavesdropper relay-link SNR. Since ZFB projects the information signal into Eve’s null space, changes in Eve’s channel quality have minimal impact on the secrecy performance. By contrast, NAN’s SOP was found to degrade significantly as the eavesdropper relay-link SNR increased, highlighting a key vulnerability of the NAN scheme in strong eavesdropper channel conditions.
- ZFB achieved a higher peak effective secrecy throughput compared to NAN across the evaluated antenna configurations. The performance gap between the two schemes widened as the number of relay antennas increased, confirming that ZFB benefits more from additional antenna resources than NAN.
- Both schemes were found to perform poorly when the eavesdropper is located in close proximity to the base station, as neither ZFB nor NAN can suppress the direct BS-to-Eve link. This represents a shared limitation that would require source-level countermeasures to address effectively.
- The derived closed-form SOP expressions were validated against simulation results and found to be in close agreement, confirming the accuracy of the analytical framework. This demonstrates that the derived expressions are reliable tools for system design and performance prediction without the need for extensive simulation.

6.3 Recommendations

Based on the conclusions above, the following recommendations are made for practitioners designing secure cooperative sensor networks:

- Deploy TAS-MRC-ZFB whenever the relay can estimate or approximate the eavesdropper’s channel direction. Even coarse directional information enables effective null-space projection.
- Target relay antenna counts of $M \geq 3$ to realise meaningful secrecy diversity gains. Below $M = 3$, the performance of both schemes converges and the benefit of additional hardware is reduced.
- Configure the system’s target secrecy rate near the optimal R_s^* to maximise effective secrecy throughput. For ZFB with $M = 5$ at 10 dB SNR, $R_s^* \approx 1.8$ bps/Hz.
- Supplement relay-based cooperative jamming with source-level directional transmission in scenarios where Eve may be proximate to Alice, as relay jamming alone cannot suppress the direct link.

6.4 Limitations

The following limitations of the current study are acknowledged:

- i. The analysis assumes perfect CSI of the eavesdropper’s channel at the relay for ZFB. In practice, this requires channel estimation that may introduce errors, particularly for passive eavesdroppers.

- ii. The model assumes a single eavesdropper with a single antenna. Cooperative eavesdropping with multiple colluding antennas may require different jamming strategies.
- iii. Power allocation between information forwarding and cooperative jamming is not jointly optimised in this work. Optimal power allocation could further reduce the SOP.
- iv. The threshold-based mode-switching mechanism (activating cooperative jamming only when the relay-to-Bob channel exceeds a threshold) was identified but not optimised in this work.

6.5 Directions for Future Work

Several natural extensions of this work are proposed for future investigation:

1. **Joint power allocation optimisation:** Derive the optimal distribution of transmit power among the BS, relay forwarding, and relay jamming components that minimises the SOP for both ZFB and NAN, and quantify the additional secrecy gains achievable through power optimisation.
2. **Optimal mode-switching threshold:** Derive analytically the optimal BS-to-relay channel quality threshold at which to activate cooperative jamming, revealing any qualitative differences in the threshold behaviours of ZFB and NAN.
3. **Imperfect eavesdropper CSI:** Extend the ZFB analysis to the case where the relay has only statistical or estimated eavesdropper CSI, characterising the SOP degradation as a function of channel estimation error variance.
4. **Multiple eavesdroppers:** Extend the system model to multiple passive eavesdroppers with potentially colluding antennas, requiring modifications to the cooperative jamming strategy.
5. **Full-duplex relay operation:** Consider full-duplex relay operation, which eliminates the half-duplex spectral efficiency loss but introduces self-interference that must be cancelled. Characterise the net security benefit of full-duplex operation for both jamming schemes.
6. **RIS-assisted cooperative networks:** Integrate reconfigurable intelligent surfaces (RIS) into the cooperative network to further enhance the legitimate channel quality and improve the robustness of cooperative jamming to Eve's proximity.
7. **Machine learning-based eavesdropper localisation:** Investigate data-driven approaches to estimating the eavesdropper's channel direction from passive monitoring of the wireless environment, enabling ZFB deployment even in nominally passive eavesdropping scenarios.

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APPENDIX A: MATHEMATICAL DERIVATION DETAILS

A.1 Proof of the ZFB Eavesdropper PDF

This appendix provides additional detail on the derivation of the eavesdropper PDF under the ZFB scheme, Equation (4.1) and Equation (4.2).

Under ZFB, the relay's beamforming vector $\mathbf{w}_2^{\text{ZFB}}$ lies in the null space of $\mathbf{h}_{RE}$, so Eve receives no information signal from the relay. Her SNR is entirely due to the direct BS-to-Eve path:

$$\gamma_E^{\text{ZFB}} = \frac{P_A d_{AE}^{-\alpha}}{N_0} |h_{AE}|^2 = \bar{\gamma}'_{AE} \cdot X_{AE}, \quad (\text{A.1})$$

where $X_{AE} = |h_{AE}|^2 / \Omega_{AE} \sim \text{Exp}(1)$ and $\bar{\gamma}'_{AE} = P_A \Omega_{AE} d_{AE}^{-\alpha} / N_0$. However, the beamforming design also uses $M - 1$ degrees of freedom at the relay for null-space projection (nulling the eavesdropper requires projecting out one dimension), and the residual $M - 1$ dimensions contribute to apparent interference terms in the SOP integral. The combined effect, after the projection geometry of (3.16) is applied and the composite random variable is characterised, yields the PDF of Equation (4.1) following the approach of Yang et al. [24].

A.2 Proof of the Auxiliary Function Φ

The auxiliary function $\Phi(p, q, \beta, \alpha)$ defined in Equation (4.10) arises from the evaluation of integrals of the form:

$$I = \int_0^\infty y^p (1 + \alpha y)^{-q} e^{-\beta y} dy. \quad (\text{A.2})$$

Applying the substitution $u = \beta y + \beta / \alpha$ and using the series expansion of $(1 + \alpha y)^{-q}$ along with the identity for the incomplete gamma function $\Gamma(n, x) = (n - 1)! e^{-x} \sum_{k=0}^{n-1} x^k / k!$ for integer n , the integral can be expressed as a finite sum of incomplete gamma functions, yielding the closed-form expression (4.10). The derivation follows Lemma 1 of [24] for the general case of real-valued parameters.

APPENDIX B: SIMULATION PARAMETER SUMMARY

Table B.1 summarises all simulation parameters used in the numerical results of Chapter 5.

Table B.1: *Summary of simulation parameters.*

Parameter	Symbol	Value
Path-loss exponent	α	3
BS-to-relay distance	d_{AR}	1.0 (normalised)
BS-to-Eve distance	d_{AE}	1.5 (Fig. 5.1); 0.8/1.5 (Fig. 5.2)
Relay-to-Bob distance	d_{RB}	1.0 (normalised)
Relay-to-Eve distance	d_{RE}	0.8
Reference BS SNR	$\bar{\gamma}_{AE}$ (before path loss)	10 dB
Reference relay SNR	$\bar{\gamma}_{RE}$ (before path loss)	10 dB
Target secrecy rate	R_s	0.5 bps/Hz (unless varied)
Relay antenna count	M	{2, 3, 5}
Transmit SNR range	$\bar{\gamma}_{AR}$	0–40 dB (unless fixed at 10 dB)

APPENDIX C: GLOSSARY OF KEY TERMS

Amplify-and-Forward (AF)

A relay protocol in which the relay amplifies the received signal without decoding and retransmits it to the destination.

Cooperative Jamming

A technique in which a relay or helper node deliberately transmits interference to degrade the quality of reception at an eavesdropper.

Diversity Order

The rate at which the outage probability decreases as a function of SNR in the high-SNR regime; a higher diversity order indicates a steeper decline in outage probability.

Effective Secrecy Throughput

The product of the target secrecy rate and the probability of not experiencing a secrecy outage; measures the reliably deliverable secret information rate.

Incomplete Gamma Function

The function $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$, a generalisation of the factorial function used in probability distributions and performance analysis.

Maximal Ratio Combining (MRC)

A diversity combining technique that coherently sums received signal copies with weights proportional to their signal amplitudes and inversely proportional to their noise levels, maximising the SNR.

Null Space

For a matrix $\mathbf{A}$, the null space is the set of all vectors $\mathbf{v}$ such that $\mathbf{A}\mathbf{v} = \mathbf{0}$.

Order Statistics

The statistics of the maximum, minimum, or k -th largest of a set of independent random variables; used to characterise TAS selection.

Physical Layer Security

Security achieved by exploiting the physical properties of the wireless channel (randomness, spatial selectivity) rather than computational cryptographic methods.

Rayleigh Fading

A statistical model for wireless fading in environments with rich scattering and no dominant line-of-sight path; the magnitude of the channel coefficient follows a Rayleigh distribution.

Secrecy Capacity

The maximum rate at which information can be transmitted with perfect secrecy; equals $[\log_2(1 + \gamma_B) - \log_2(1 + \gamma_E)]^+ / 2$ for half-duplex relaying.

Secrecy Outage Probability

The probability that the instantaneous secrecy capacity falls below a prescribed target secrecy rate; the primary PLS performance metric in fading channels.

Transmit Antenna Selection (TAS)

A spatial multiplexing technique in which a single transmit antenna is selected from M available antennas to maximise the received SNR at the legitimate user.

Zero-Forcing Beamforming (ZFB)

A relay beamforming strategy that projects the information signal into the null space of the eavesdropper's channel, completely eliminating the relay-to-eavesdropper information leakage.